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Anonymized Network Sensing Using C++26 `std::execution` on GPUs

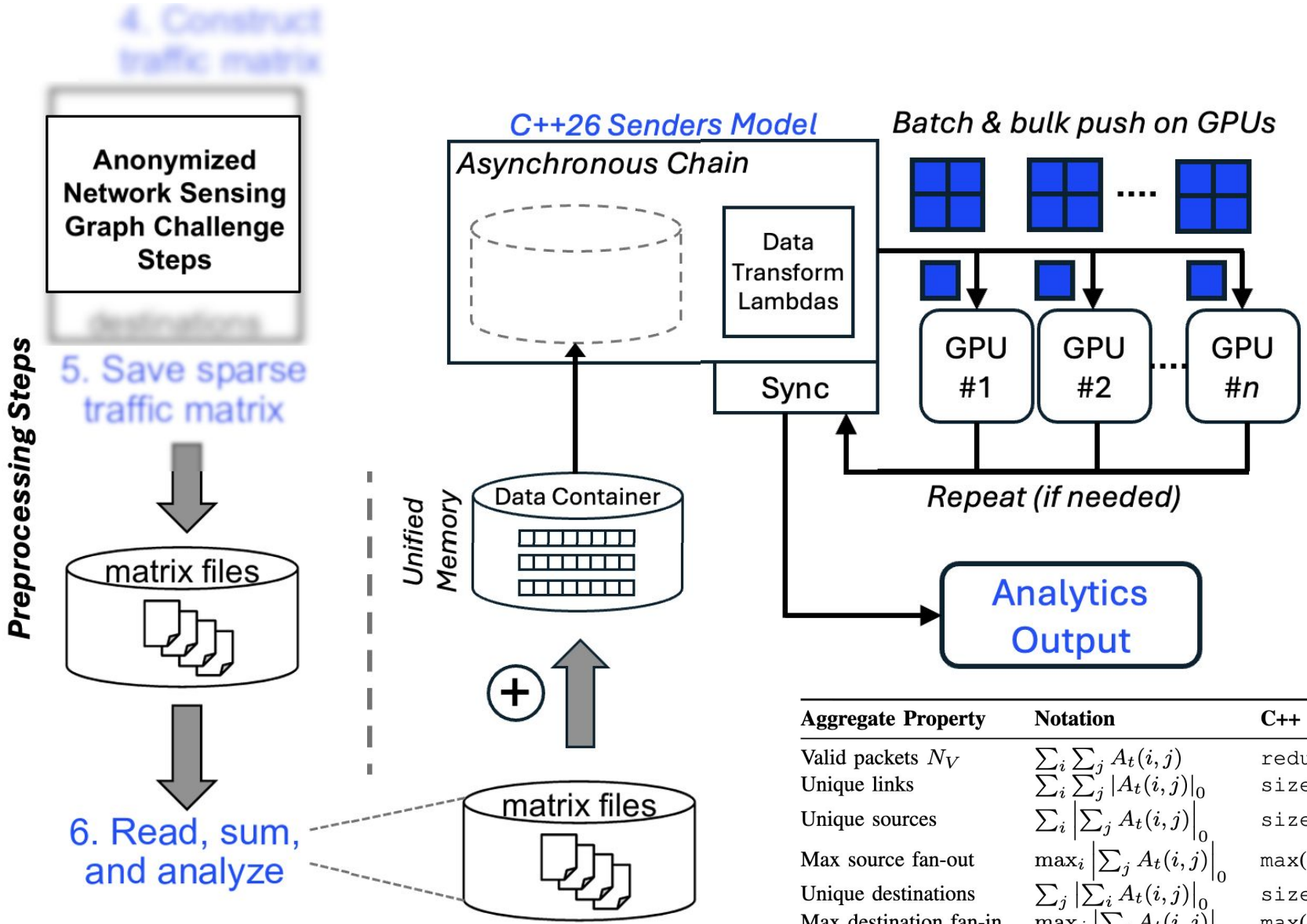
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2025 High Performance Extreme Computing:
Anonymized Network Sensing Graph Challenge

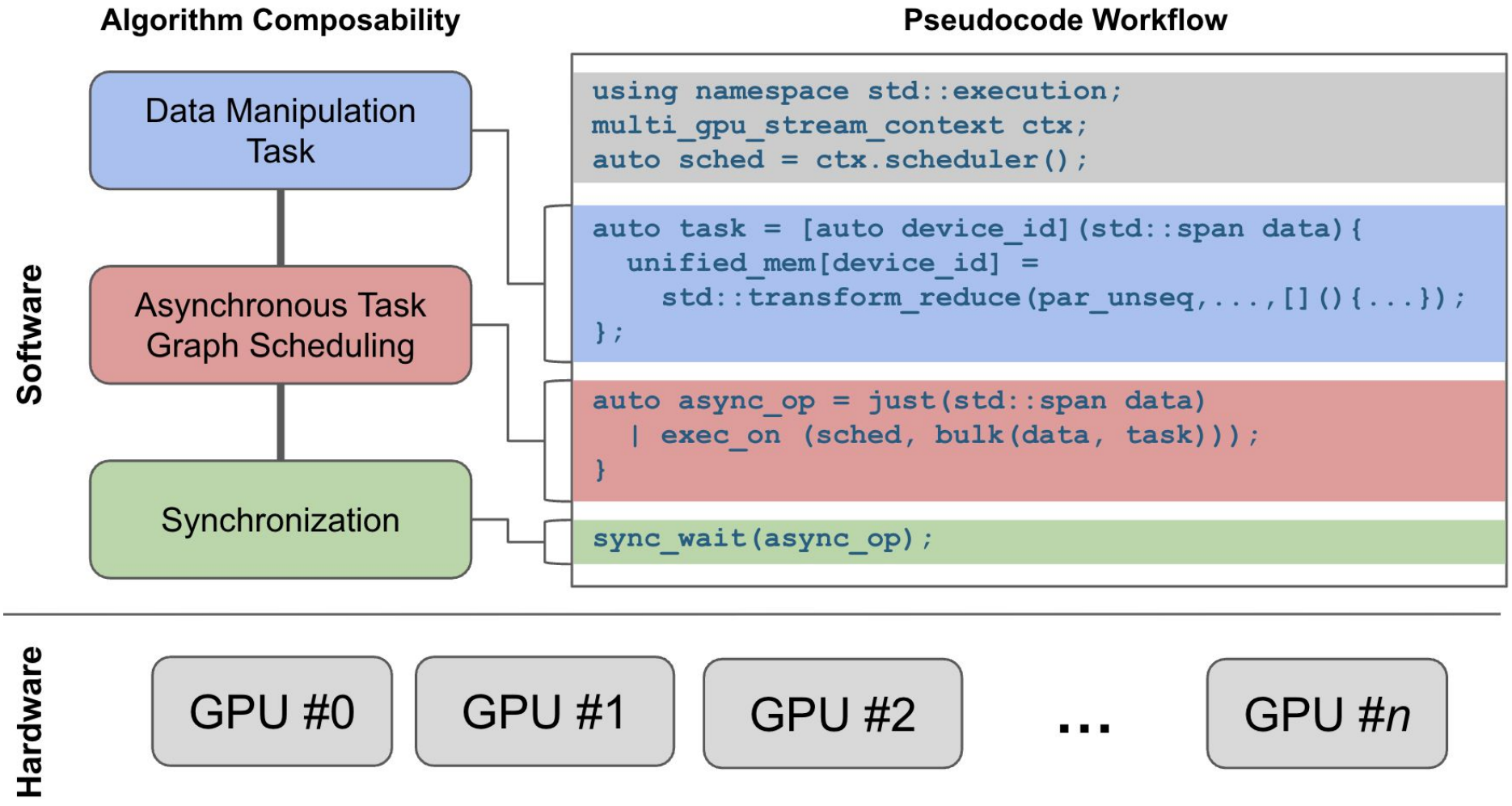
September 17, 2025

Anonymized Network Sensing Graph Challenge – Traffic Matrix Analysis

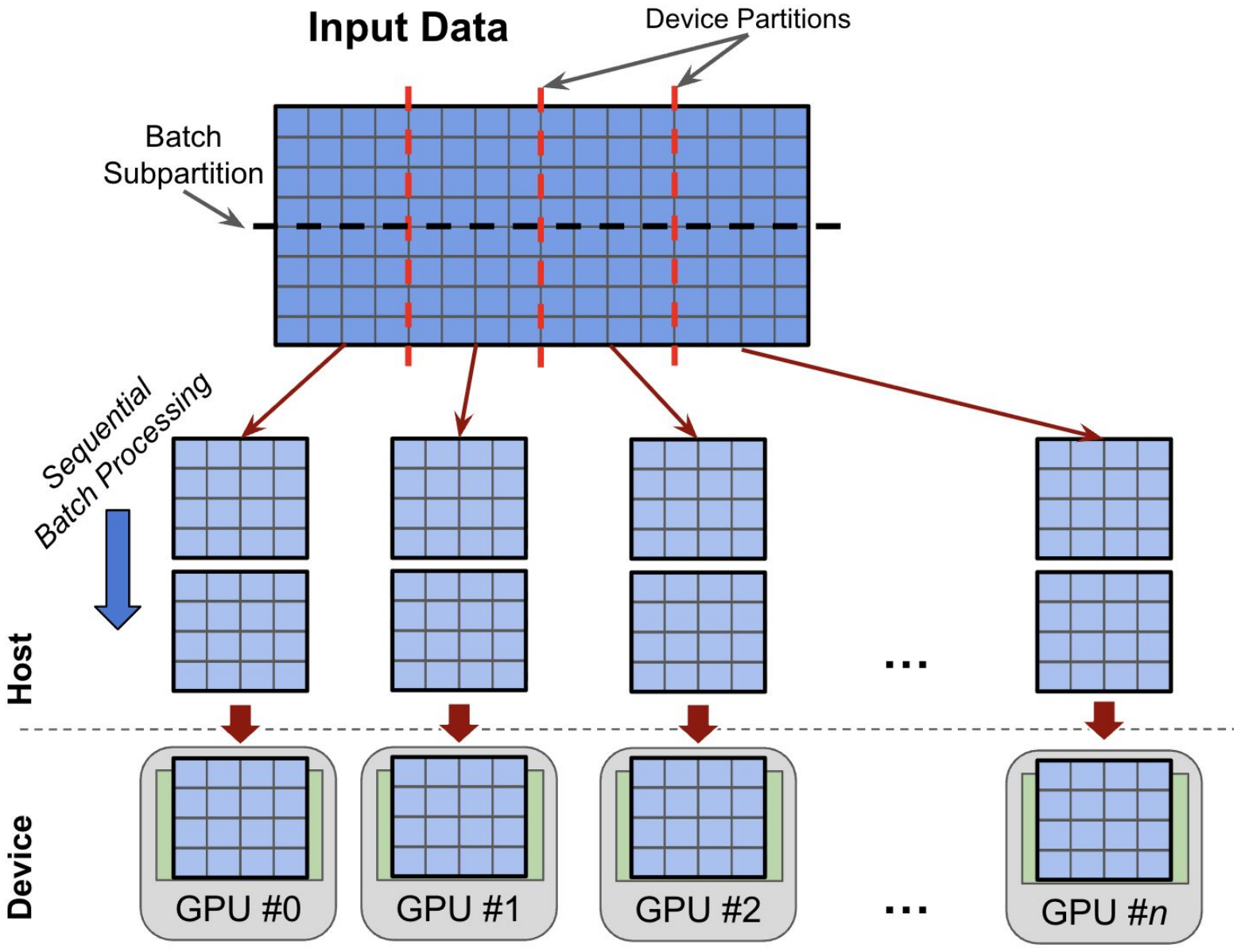


Aggregate Property	Notation	C++ function
Valid packets N_V	$\sum_i \sum_j A_t(i,j)$	`reduce(`*weights*`)`
Unique links	$\sum_i \sum_j \lvert A_t(i,j)\rvert_0$	`size(`*edges*`)`
Unique sources	$\sum_i \lvert \sum_j A_t(i,j)\rvert_0$	`size(`*row_sums*`)`
Max source fan-out	$\max_i \lvert \sum_j A_t(i,j)\rvert_0$	`max(`*degrees*`)`
Unique destinations	$\sum_j \lvert \sum_i A_t(i,j)\rvert_0$	`size(`*col_sums*`)`
Max destination fan-in	$\max_j \lvert \sum_i A_t(i,j)\rvert_0$	`max(`*degrees*`)`

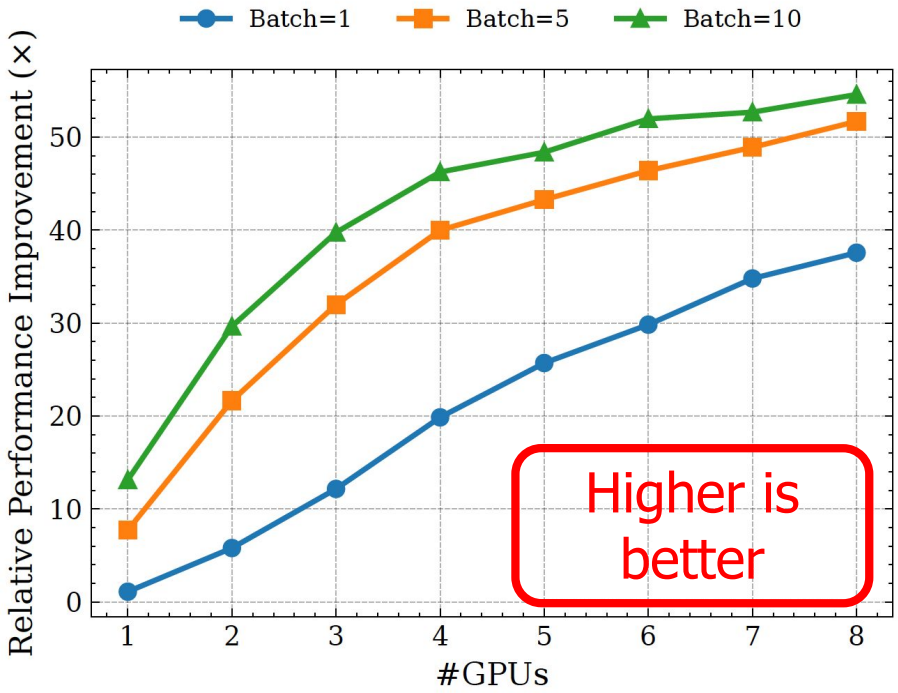
Detailing `C++26 std::execution` Senders Model



Batching (when partitions > GPU memory)



Evaluations

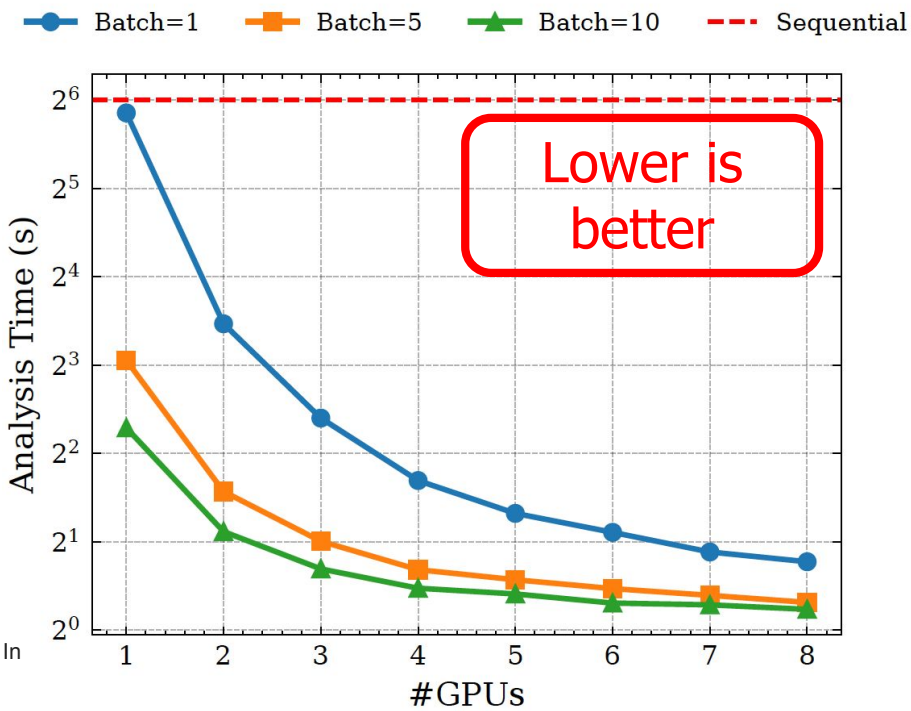


Outperform sequential using 8x A100 GPUs by 55x (2x compared with 2024 champion, *sans**)

Batching + multi-GPU scheduler improves performance

Multi-GPU performance with no CUDA device code

Workflow flexible to a variety of graph analytics



* Zhao K, Zhou Y, Pan H, Wang Z, Zhong S, Tian C. Sans: Streaming Anonymized Network Sensing. In 2024 IEEE High Performance Extreme Computing Conference (HPEC) 2024 Sep 23 (pp. 1-7).

Acknowledgements & Contact

Contact

- Michael Mandulak: mandum@rpi.edu
  - Available for HPC-related roles → 2026!
- Sayan Ghosh: sayan.ghosh@pnnl.gov

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Collaborators:

- S M Ferdous (PNNL)
- Mahantesh Halappanavar (PNNL)
- George Slota (RPI)

[code] https://github.com/mmandulak1/stdexecANSGC